Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/20/2023
PI Name:	Clark, Torin K. Ph.D.		
Project Title:	Detecting Pilot Spatial Disorientation to Trigger Active Countermeasures During Lunar Landing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80303-7813	**Congressional District:**	2
Comments:	NOTE: PI moved to University of Colorado after NSBRI Postdoctoral Fellowship concluded in late 2015 (Ed., 9/1/17)		
Project Type:	Ground	**Solicitation / Funding Source:**	2020-2021 HERO 80JSC020N0001-HHP, OMNIBUS3 Human Research Program: Human Health & Performance Appendix E; Omnibus3-Appendix F
Start Date:	12/23/2022	**End Date:**	12/22/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Added Dr. Jordan Dixon (Charles Stark Draper Laboratory, Inc.) as a Co-Investigator for his expertise in the spatial disorientation (SD) triggering algorithm. Change occurred September 28, 2023.		
COI Name (Institution):	Holder, Sherrie Ph.D. (Charles Stark Draper Laboratory Inc) Endsley, Tristan Ph.D. (Charles Stark Draper Laboratory Inc) Vance, Eric Ph.D. (University of Colorado, Boulder) Dixon, Jordan (Charles Stark Draper Laboratory, Inc.)		
Grant/Contract No.:	80NSSC23K0449		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	During transit in microgravity, astronauts will reinterpret neurovestibular stimuli, prior to initial exposure to partial gravity when landing on the Moon or Mars. This poses a risk of spatial disorientation and impaired manual control performance during piloted planetary landings. Here, we propose to develop, validate, and assess a system for detecting when astronauts may become disoriented in real-time, such that it can be used to trigger active countermeasures for piloted planetary landings. Our approach leverages a well-validated computational model for human spatial orientation, now applied to partial gravity planetary landings. Incorporating microgravity neurovestibular adaptation, the vehicle motions of each landing trajectory are processed in real-time by the computational model to detect pilot spatial disorientation. We will assess the system using a ground-based lunar landing analog, combining a gravity transition (3 Gx) with a motion-based planetary landing simulation. First, we will experimentally tune and re-validate the computational model for detecting spatial disorientation, accounting for the effects of the recent gravity transition. Then, using the high-fidelity Disorientation Research Device, we will assess the benefit of the active countermeasure triggering system. Critically, this approach of triggering manual control countermeasures only when they are needed (i.e., when the pilot is about to be disoriented) avoids the added burden on the pilot to continuously process additional sensory information or otherwise have increased workload. We aim to deliver a validated performance support tool for triggering active countermeasures for pilot spatial disorientation during manually controlled lunar landings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing a means to estimate a human's perception of self-orientation and the potential for spatial disorientation in real-time, in order to trigger an intervention to enhance performance and safety. In addition to the application of piloted lunar landing, our approach can be applied to pilots of terrestrial aircraft or other vehicles in which spatial orientation perception is critical. Finally, our enhancement of models for human spatial orientation perception applies to humans more generally, such as high performance performance athletes (e.g., gymnasts), critical operators (e.g., scuba divers), or patient populations (e.g., mTBI patients of older adults at risk for impaired balance).
Task Progress:	This is the first year of the project, dedicated to project definition, so limited experimental progress was made. However, technical and administrative progress was made on a number of fronts, which is summarized here. First, the team completed a kickoff meeting, defining short and long term objectives, administrative and technical roles and responsibilities, program management approaches, and addressing outstanding issues from the proposal. In addition, the team coordinated with colleagues and collaborators relevant for this project. This included kickoff meetings, telecons, and in-person meetings with NASA Human Research Program personnel, scientists, and managers. Specifically, we coordinated programmatic schedules and availability for planned experiments using the Navy Aeromedical Research Unit-Dayton's (NAMRU-D's) Disorientation Research Device (DRD, aka the Kraken). As a major activity, team personnel attended and participated in a multi-day meeting at NASA Langley Research Center. Activities included sharing of technical information and demonstrations of prototype displays, interfaces, and control modes planned for piloted lunar landing as part of the Artemis program. The team presented to NASA colleagues within the Human Research Program, crew training, and flight simulation groups, regarding the proposed approached, planned experiments, and deliverables of this project. Feedback from NASA colleagues was documented and is being integrated into our project plans. Finally, our team has coordinated with colleagues at the NAMRU-D regarding initial plans for the capstone experiments using the DRD. This has included technical discussions of necessary instrumentation (e.g., accelerometers and inertial measurement units) for our planned experiments, as well as administrative coordination (e.g., developing Cooperative Research and Development Agreement / CRADA documentation). Through coordination with NASA Human Research Program personnel and managers, our team was tasked with modifying our original proposal to more extensively and precisely defining the countermeasure intervention we envisioned implementing and assessing, that would be triggered by our spatial disorientation algorithm. Through considering various alternatives, either that our team has previous experience with, or those which have been proposed by others, we refined the potential alternatives. Our plan, the modification of the original proposal, and its resubmission to the NASA Human Research Program is nearly complete. At this time, we envision the primary approach is for an adaptive display, which enhances the saliency (visual and auditory) of vehicle attitude and motion information in real-time based upon the spatial disorientation algorithm. Specifically, only when the algorithm deems the astronaut pilot is likely to be spatially disoriented does it automatically increase the saliency of critical information on the instrumentation display regarding vehicle attitude (roll and pitch), and motion, to help the pilots reorient themselves in order to improve piloting performance and safety. When the algorithm estimates the pilot is not suffering from spatial disorientation, the nominal instrument display information (including landing point select, fuel remaining, terrain hazards, etc.) will be provided, avoiding an unnecessary burden upon the pilot. As such, we envision the adaptive display system will serve as a pilot aid, assisting the crewmember as needed. In addition, we are considering a few alternative countermeasure interventions and envision initial human-in-the-loop evaluation in our laboratory, prior to final selection for the use in NAMRU's DRD lunar landing simulator. In addition to making progress in preparation for capstone experiments in the DRD, we are making technical progress for preliminary experiments in our laboratory. First, we have developed a laboratory based capability for the hyper-Gx paradigm, planned for eventual use in the DRD, in a gravity transition analog. Participants spin on a centrifuge in hyper-Gx (i.e., the net force is "into the chest") for an extended period of time. When the centrifuge is spun down, the transition back from hyper-Gx tends to lead to misperception of orientation, impaired sensorimotor function, and motion sickness, serving as an analog for astronaut gravity transitions. In our laboratory, the subject is positioned 9 feet off-axis and spun to produce 2Gx for approximately 1 hour. Thus far, we have safety tested and assessed the hyper-Gx analog in a number of participants, demonstrating feasibility. In addition, we have made advancements to our human-rated motion device (the Tilt-Translation Sled, or TTS) which we plan to use for preliminary investigations prior to the capstone assessments in the DRD. Specifically, we have implemented a manual control piloting mode that is analogous to lunar landing, whereby joystick deflections lead to a roll tilt motion, which is then coupled to a lateral translation in the same direction (i.e., tilt to the left leads to translation to the left). The subject is tasked with controlling translation motions via this coupled motion, similar to the manual control task during the final stages of lunar landing prior to touchdown. This control mode has been fully implemented and tested with pilot subjects in the loop. Finally, as an initial demonstration, we have created a software implementation of our preliminary spatial disorientation algorithm on the TTS device. In real-time, the algorithm processes TTS motions (roll tilt and lateral translation) through our computational model for human spatial orientation perception and then computes a unidimensional metric of how the pilot is likely to be disoriented at that instant in time.

	The spatial disorientation metric is programmed on the TTS to trigger a change in the instrument display shown on a tablet in front of the participants. Initial pilot tests have shown that when the spatial disorientation metric triggers the instrument display, the pilot is better able to pilot the manual control motions of the TTS. Future work will build upon this to perform a preliminary human subject experiment, leading up to the capstone DRD experiment.
Bibliography Type:	Description: (Last Updated: 11/28/2025)
Abstracts for Journals and Proceedings	Dixon JB, Endsley T, Clark TK. "Novel methodology and experimental ratings for real-time computational detection of pilot spatial disorientation." Vestibular Oriented Research Meeting, Boulder, Colorado, June 25-29, 2023. Abstracts. Vestibular-Oriented Research Meeting, Boulder, Colorado, June 25-29, 2023. J Vestib Res. 2023 Aug 14;33(4):231-78. https://doi.org/10.3233/VES-230300 ; PMID: 37355918 , Aug-2023